

# Programmatic Access to the Code of Life

An overview of the AlphaGenome API and Model Architecture



```
class AlphaGenomeModel(BaseModel):
    def __init__(self, config):
        self.encoder = Encoder(config.encoder_layers)
        self.decoder = Decoder(config.decoder_layers)

    def forward(self, dna_sequence, metadata):
        encoded_sequence = self.encoder(dna_sequence)
        decoded_output = self.decoder(encoded_sequence, metadata)
        return decoded_output

{
  "sample_id": "AG-1024",
  "sequence": "ATGCGTACGTTAGC...",
  "metadata": {
    "species": "Homo sapiens",
    "chromosome": "chr1",
    "start_pos": 123456,
    "end_pos": 789012
  }
}

[0.023, 0.567, 0.112, 0.890, ...],

{ "gene_expression": {
  "BRCA1": 4.5,
  "TP53": 3.2
},
  "variant_calls": [...]
}
```



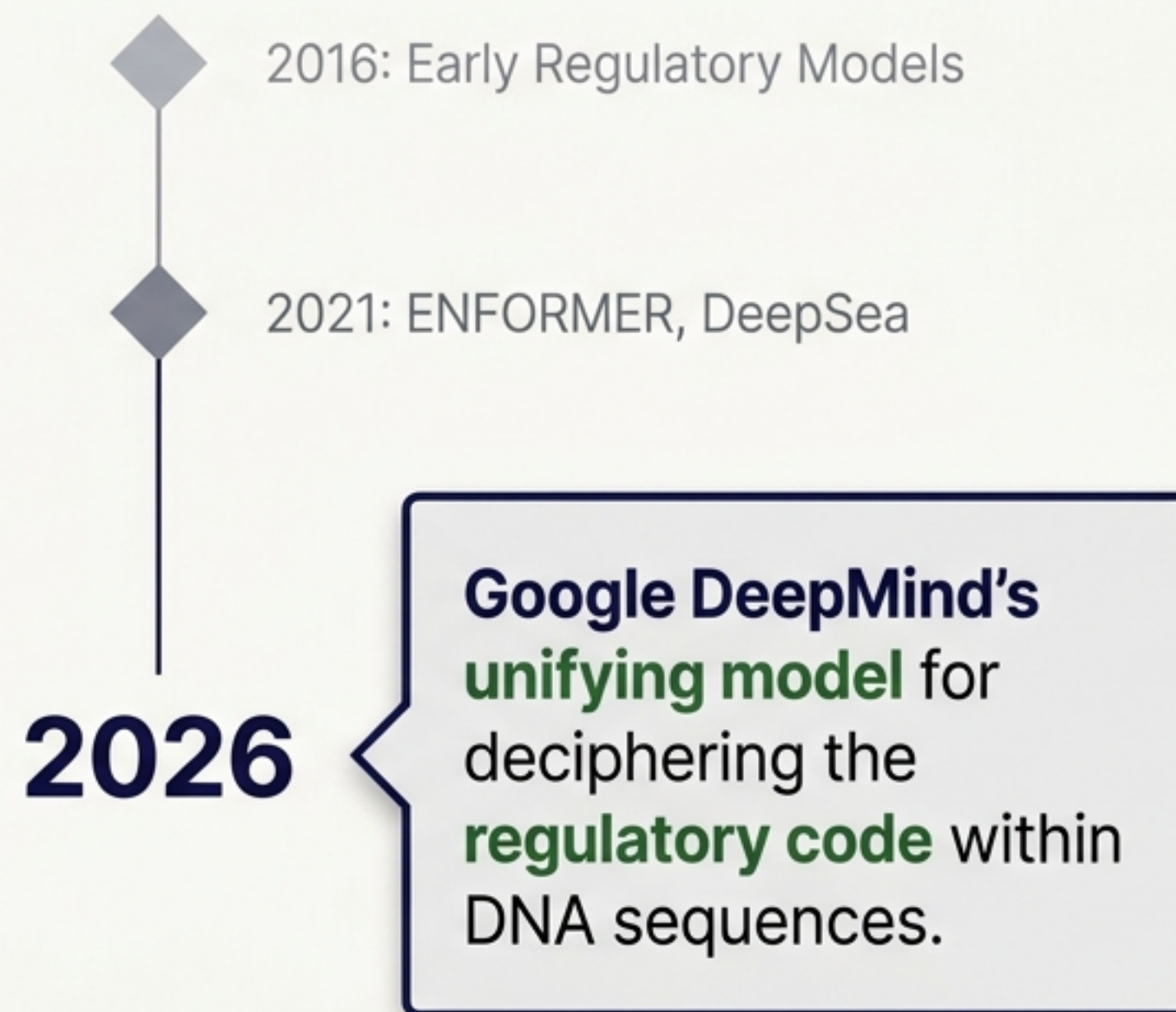
# The State-of-the-Art in Variant Effect Prediction

## Advancing regulatory variant effect prediction with AlphaGenome

**Authors:** Ziga Avsec, Natasha Latysheva, Jun Cheng, et al.

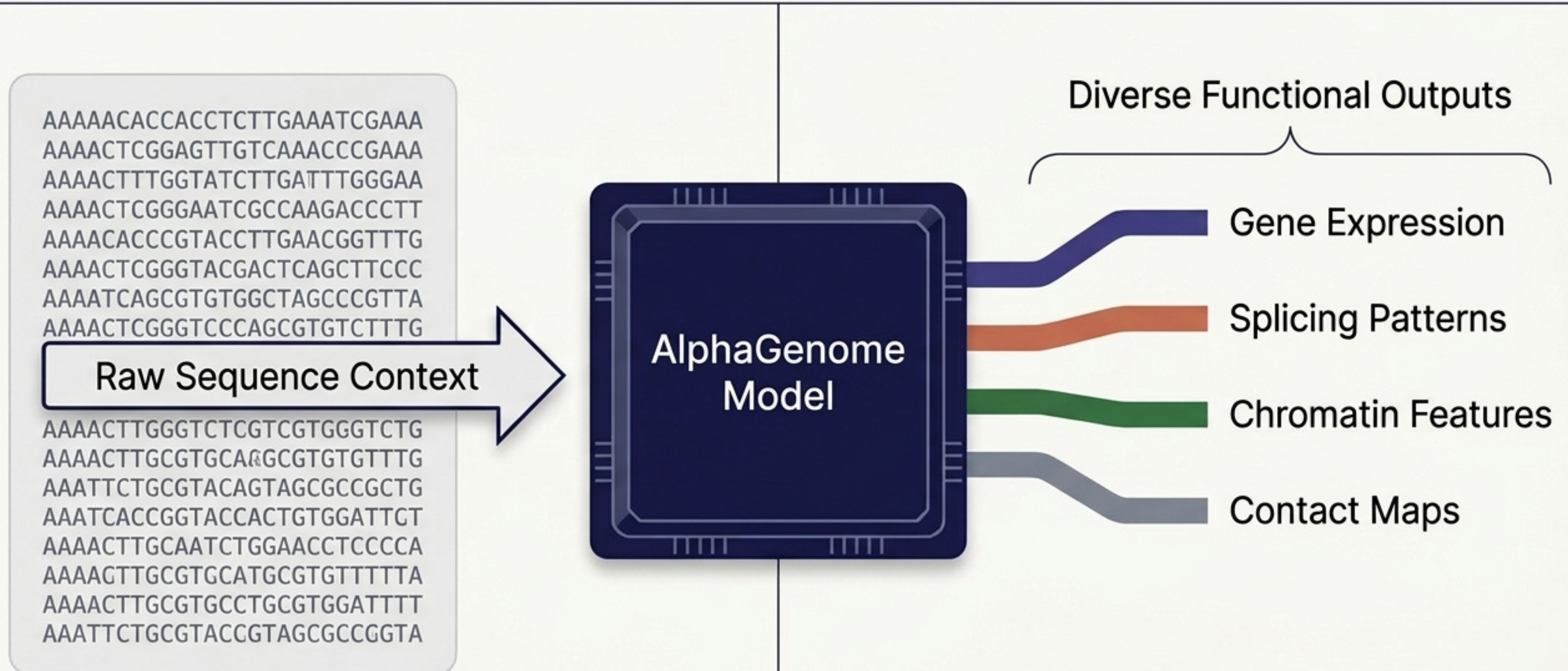
**Journal:** Nature (Vol. 649, Issue 8099)

**DOI:** 10.1038/s41586-025-10014-0



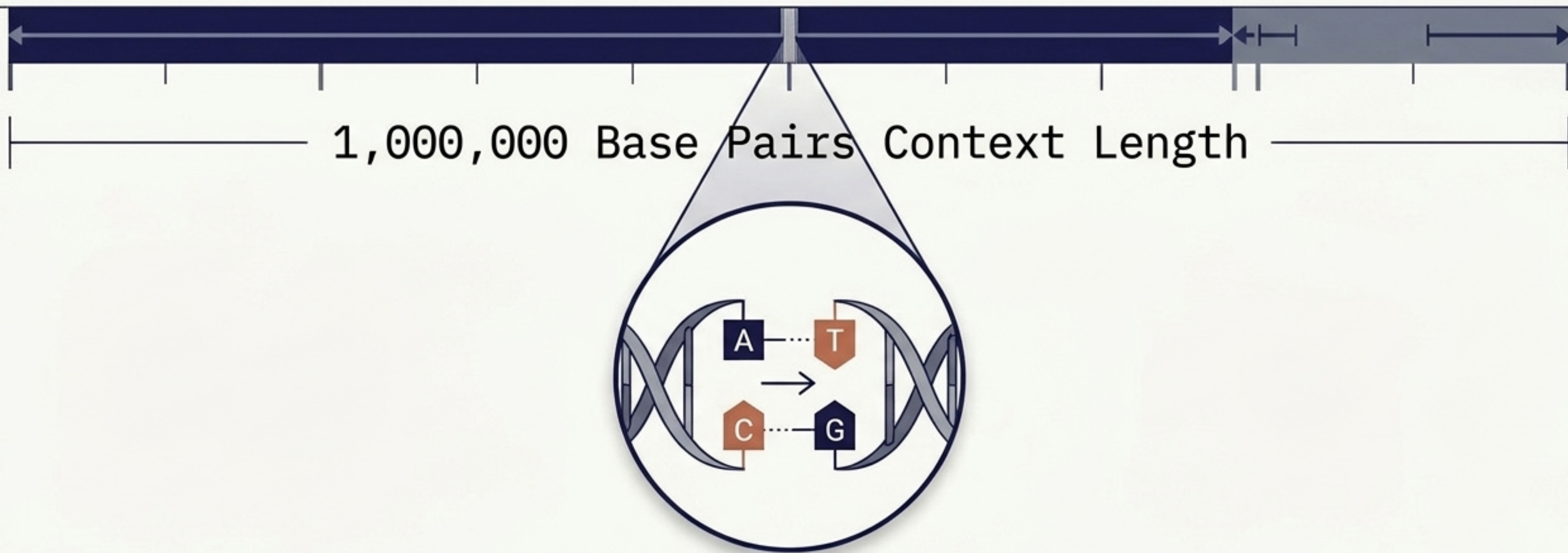


# Decoding Regulatory Complexity at Scale





# Unprecedented Context. Pinpoint Accuracy.

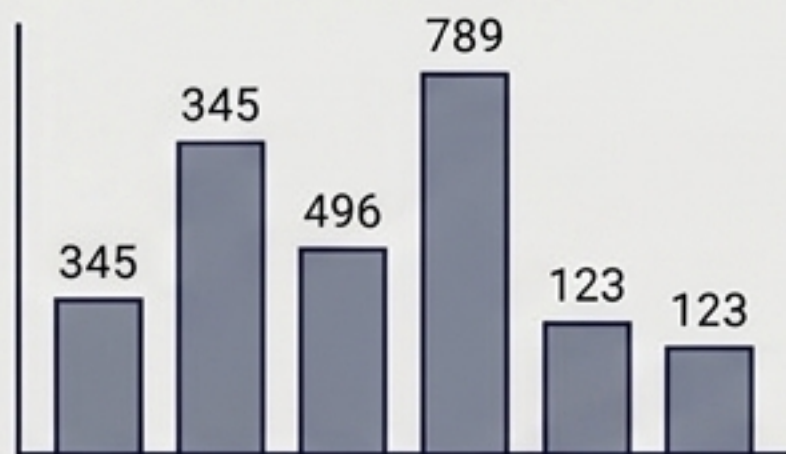


## Single Base-Pair Resolution

Analyze macroscopic genomic regions without sacrificing microscopic precision.

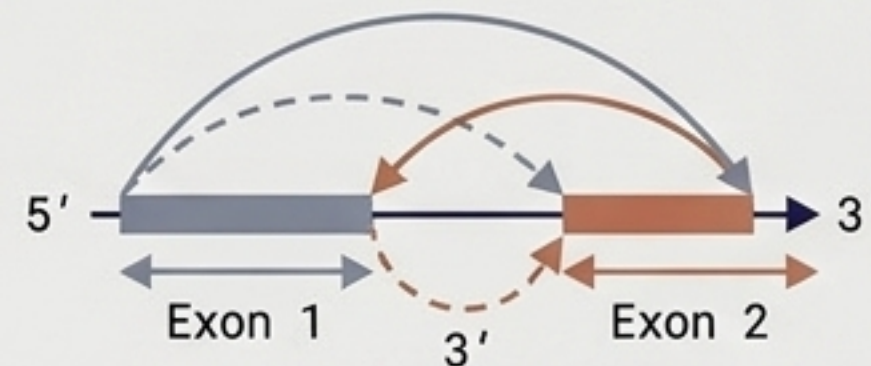


# The Four Pillars of Prediction

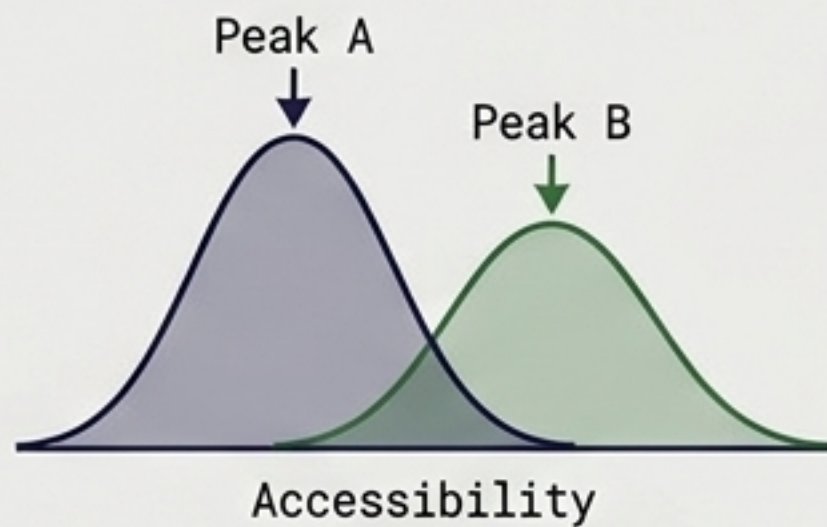


Gene  
Expression

Splicing  
Patterns

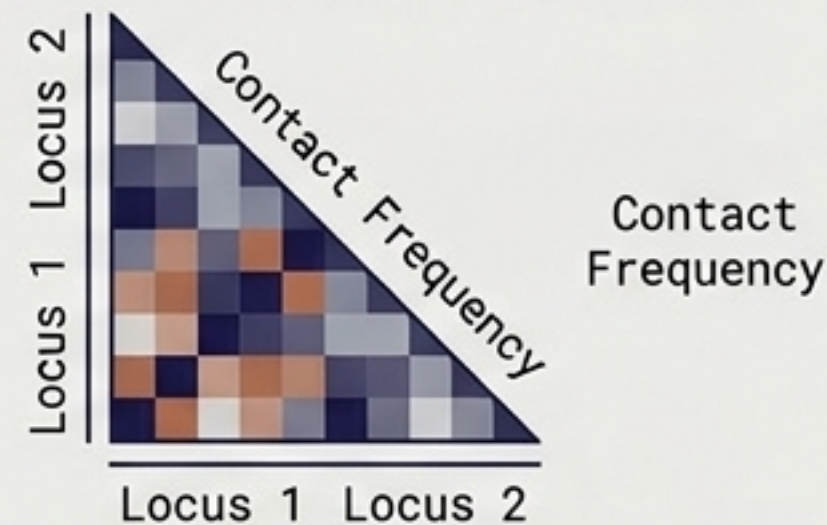


Multimodal  
Outputs



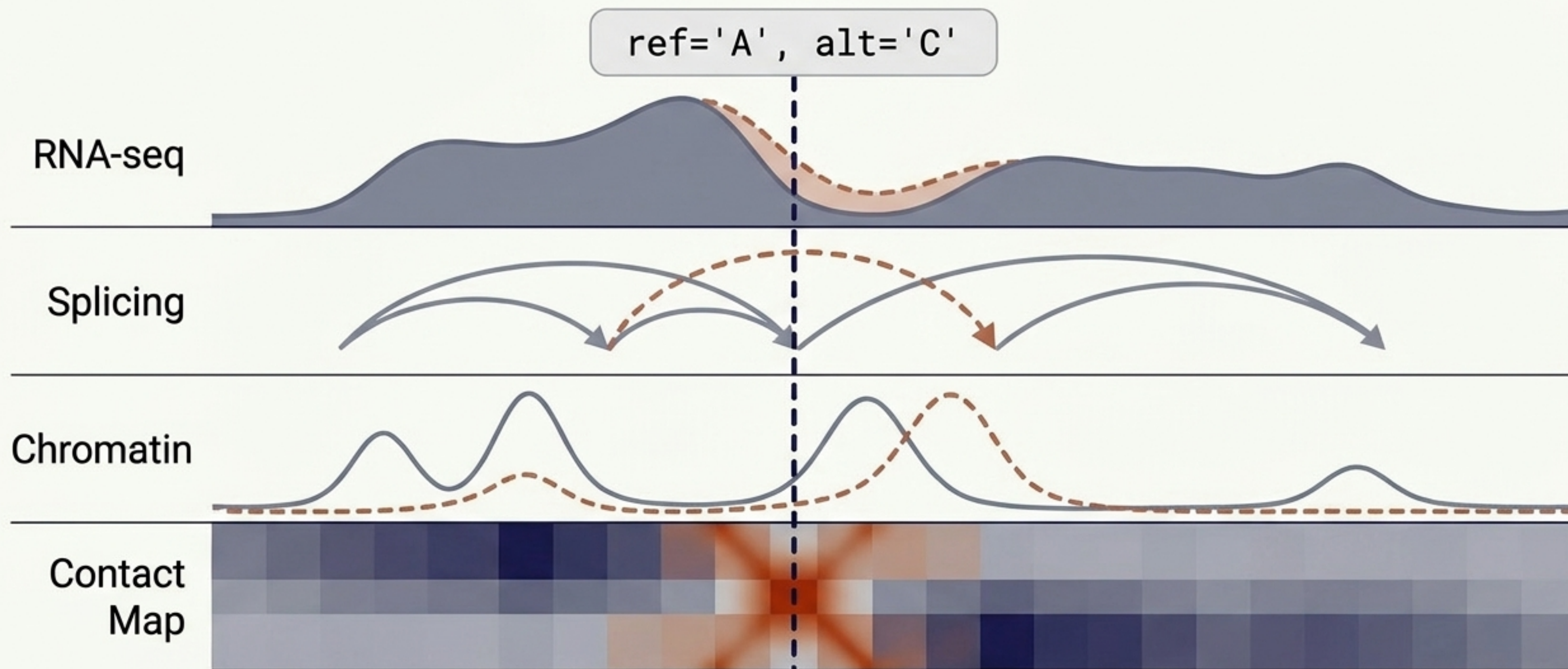
Chromatin  
Features

Contact  
Maps





# The Unified Regulatory Dashboard





# Two Paths to Discovery

## AlphaGenome API

|         |                                                  |
|---------|--------------------------------------------------|
| Format  | Dynamic, On-Demand<br>Python API                 |
| Scope   | Targeted analyses,<br>custom genomic regions     |
| Scale   | Thousands of predictions<br>(Not suited for >1M) |
| Outputs | Deep multimodal<br>visualization tracks          |

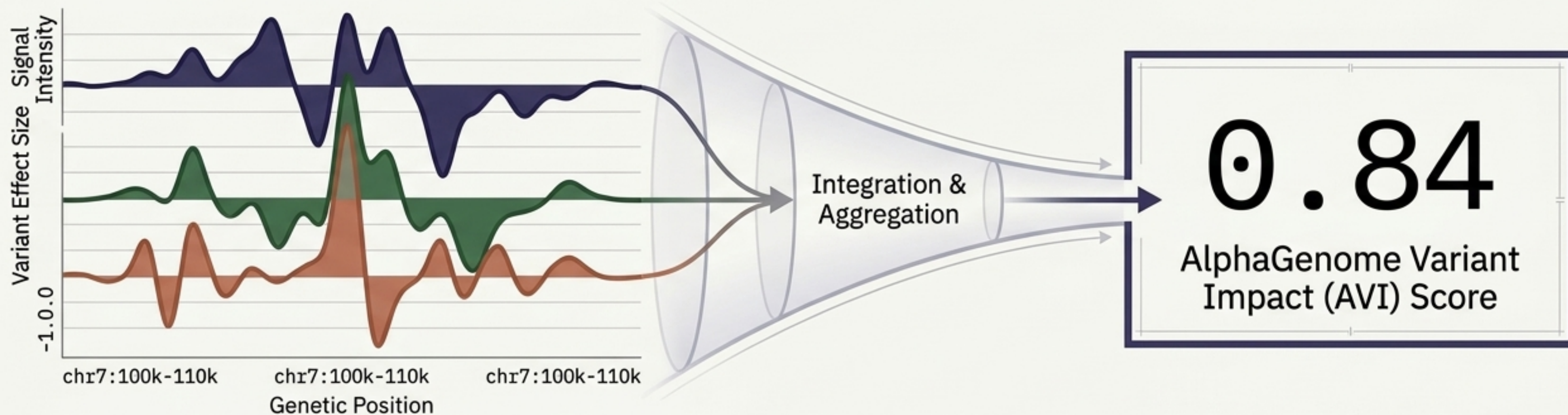
## AlphaGenome Atlas

|         |                                           |
|---------|-------------------------------------------|
| Format  | Static, Pre-computed<br>resource          |
| Scope   | The entire human genome                   |
| Scale   | High query rates,<br>whole-genome queries |
| Outputs | AVI Scores & Feature<br>Importances       |

Use the API for deep, custom mechanistic analysis.  
Use the Atlas for rapid, genome-wide variant scoring.



# Distilling Complexity: The AVI Score



**Pre-computed** across the entire human genome.

**Aggregates** variant effects into a single predictive metric.

Includes corresponding **feature importances** for interpretability.

Accessible **instantly** via the AlphaGenome Atlas.



# The Anatomy of a Query

## Step 1: Authenticate

```
dna_client.create(API_KEY)
```

## Step 2: Define Context

```
genome.Interval(chr, start, end)
```

## Step 3: Introduce Mutation

```
genome.Variant(pos, ref, alt)
```

## Step 4: Request Modality

```
predict_variant(OutputType.RNA_SEQ)
```

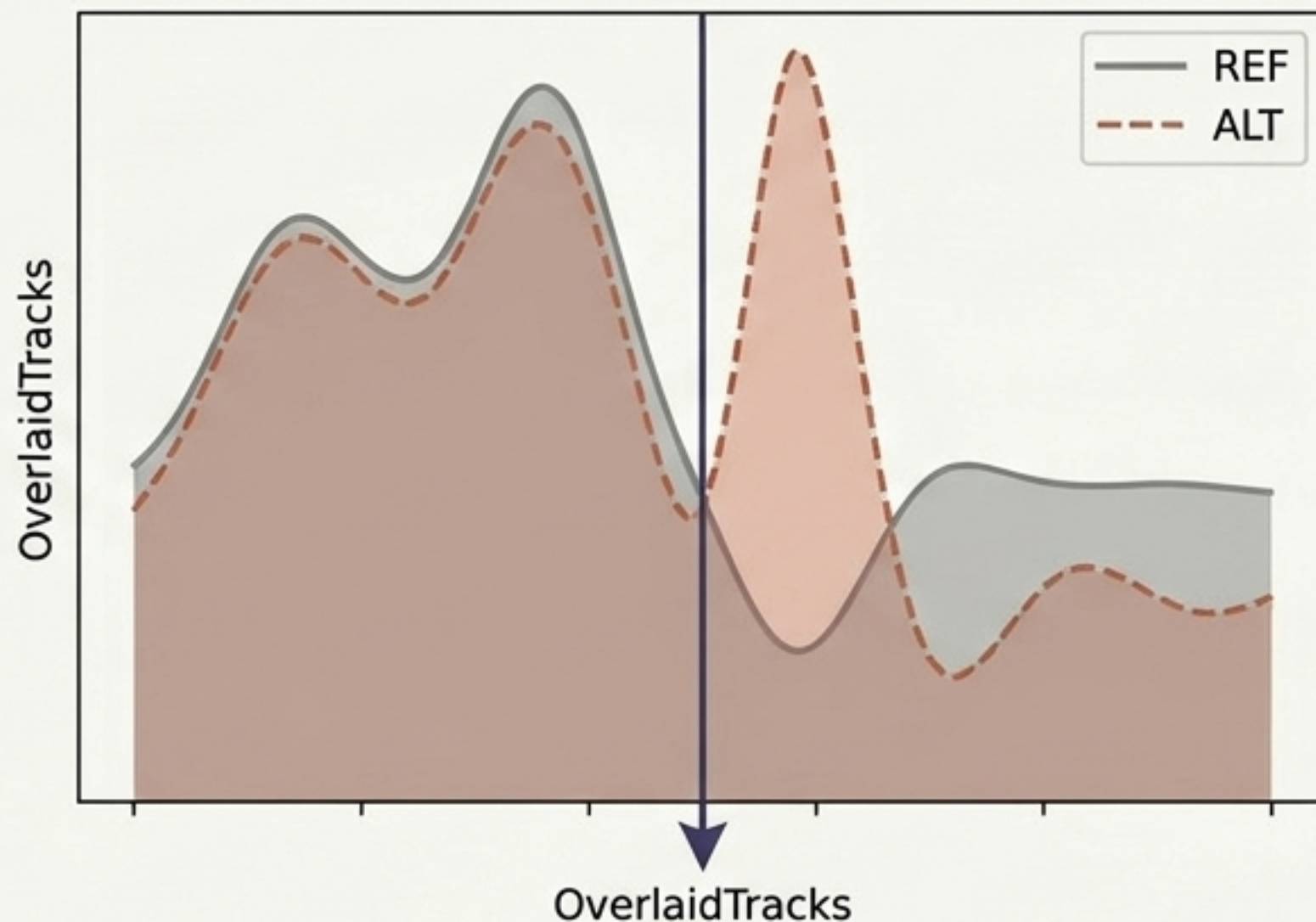
## Step 5: Visualize

```
plot_components.plot()
```



# From Variant Definition to Visual Insight

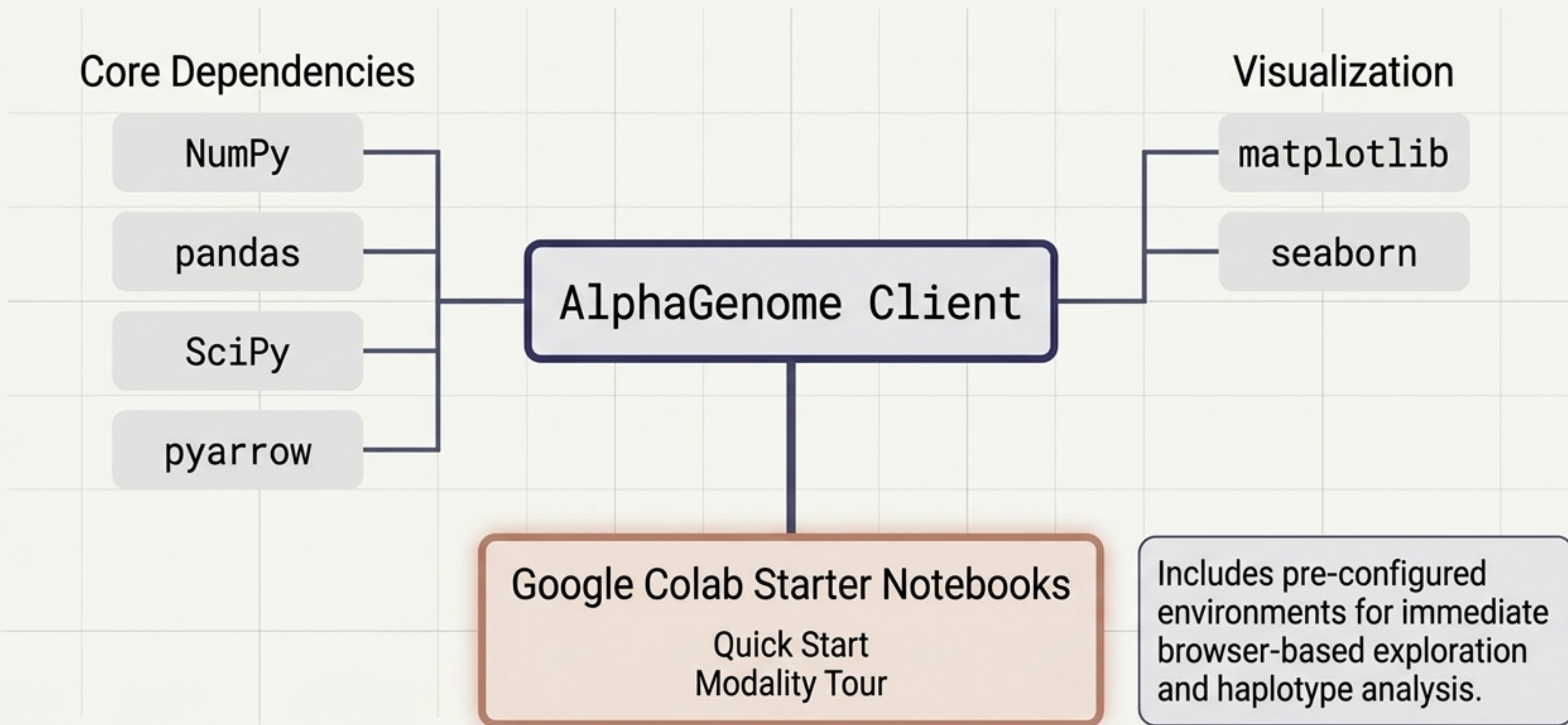
```
interval = genome.Interval(  
    chromosome='chr22', start=35677410,  
    end=36725986)  
variant = genome.Variant(  
    chromosome='chr22',  
    position=36201698,  
    reference_bases='A',  
    alternate_bases='C')  
outputs = model.predict_variant(...,  
    requested_outputs=  
    [dna_client.OutputType.RNA_SEQ])  
...
```



Generate complex multi-modal visualizations in a single script using the built-in visualization library.



# Engineered for the Scientific Python Ecosystem





# Access Parameters & Governance



## Non-Commercial Use

API is offered free of charge for non-commercial research and theoretical modeling.



## Commercial Access

Available for commercial deployment exclusively via Google Cloud integration.



## Training Restriction

Outputs and Atlas data are strictly prohibited from being used to train other machine learning models.



## Clinical Restriction

Predictions are for research only; strictly prohibited for clinical decision-making or medical advice.



# Initialization Sequence

```
> git clone https://github.com/google-deepmind/alphagenome.git  
> cd alphagenome  
> pip install .
```

Documentation:  
[alphagenomedocs.com](https://alphagenomedocs.com)

Community Forum:  
[alphagenomecommunity.com](https://alphagenomecommunity.com)

Get API Key:  
[deepmind.google.com/science/alphagenome](https://deepmind.google.com/science/alphagenome)

For general support or feature requests: [alphagenome@google.com](mailto:alphagenome@google.com)



# Acknowledging the Open Science Ecosystem

## Reference Datasets:

Incorporates a modified version of the GENCODE dataset (Copyright © 2024 EMBL-EBI).

Citation: Frankish A, et al. (2018) GENCODE reference annotation for the human and mouse genome.

## Open Source Foundations:

Built upon robust community libraries including Abseil, gRPC, JAX ecosystem (ml\_dtypes, jaxtyping), and the standard scientific Python stack.